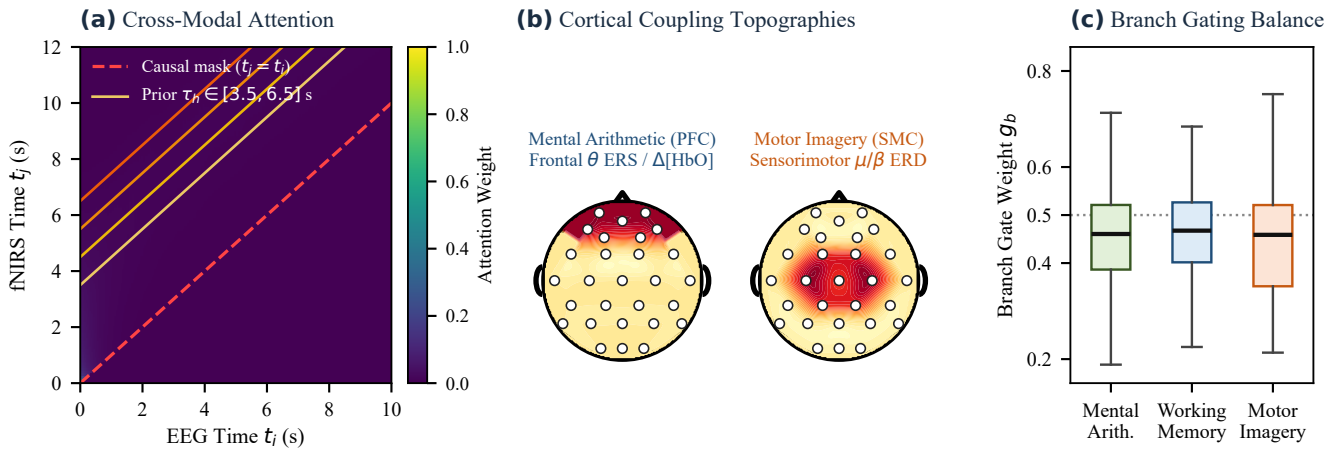

(a) Cross-Modal Attention
Causal mask ($t_j = t_i$)
Prior $\tau_n \in [3.5, 6.5]$ s
fNIRS Time t_j (s)
EEG Time t_i (s)
Attention Weight
(b) Cortical Coupling Topographies
Mental Arithmetic (PFC)
Frontal θ ERS / Δ[HbO]
Motor Imagery (SMC)
Sensorimotor μ/β ERD
(c) Branch Gating Balance
Branch Gate Weight g_b
Mental Arith.
Working Memory
Motor Imagery